

## Professional Summary

B.Tech graduate in Artificial Intelligence and Data Science with experience in designing and developing AI-driven applications. Proficient in Python, TensorFlow, and CNN architectures, with experience developing high-accuracy models for image classification, document ranking, and process automation. Experienced in data preprocessing, machine learning pipeline development, and technical communication for project evaluation and stakeholder presentations.

## Education

**Bachelor of Technology – Artificial Intelligence and Data Science**  
Karunya Institute of Technology and Sciences

**Aug 2022 - Jun 2026**  
**CGPA: 8.67**

**Higher Secondary Certificate (HSC) – Class 12**  
Geetha Matriculation Higher Secondary School, Tamil Nadu State Board

**2022**  
**Percentage: 77%**

**Secondary School Leaving Certificate (SSLC) – Class 10**  
Geetha Matriculation Higher Secondary School, Tamil Nadu State Board

**2020**  
**Percentage: 93%**

## Internship Experience

**Data Science Intern, LITZ Tech, Coimbatore**

**May 2024 - Jun 2024**

- Leveraged Python, Pandas, and NumPy to preprocess and transform **10,000+** rows of real-time data, enhancing data readiness for modelling.
- Conducted exploratory data analysis (EDA) to uncover patterns and anomalies, guiding the selection and optimization of **2** machine learning models with a **15%** boost in accuracy.
- Documented workflows via **3+** Jupyter notebooks and supported **2** stakeholder presentations for model evaluation and validation.

## Technical Skills

**Programming Languages:** Python, SQL

**AI/ML Libraries & Tools:** Pandas, NumPy, Matplotlib, TensorFlow, Keras, OpenCV

**Concepts:** Machine Learning, Deep Learning, NLP, CNNs, Data Analysis

**Web & Dev Skills:** HTML, CSS, JavaScript, Flask, RESTful APIs

**Tools & Version Control:** Git, GitHub, VS Code

## Academic Projects

**Food Image Classification and Calorie Estimation**

**May 2024 - Jun 2024**

- Engineered a CNN-based deep learning model using Python, TensorFlow, and Keras to classify **80** Indian food categories, attaining over **85%** accuracy.
- Estimated calorie values with a  $\pm 10\%$  margin of error by integrating nutritional data and mapping results to real-world food classes.
- Executed the full ML pipeline, including preprocessing, model training, hyperparameter tuning (**5+** iterations), and producing reproducible documentation.

**Optimized Document Ranking with BIM**

**Nov 2023 - Dec 2023**

- Designed and implemented a text retrieval system using TF-IDF vectorization and cosine similarity, achieving a **90%** relevance score across a **1,000+** document corpus.
- Enhanced retrieval precision by integrating the Binary Independence Model (BIM) for probabilistic relevance feedback.
- Improved query relevance by **20%** through TF-IDF weight optimization and empirical query testing.

**Face Detection Attendance System Using Python**

**Oct 2022 - Nov 2022**

- Created a real-time facial recognition system using OpenCV, reducing manual attendance workload by **90%**, capable of processing **30+** faces per minute.
- Integrated with CSV-based timestamp logging to automate attendance records and enhance operational efficiency.

## Achievements

- Secured **1st** Rank in the department in 7th Semester GPA.
- Ranked **2nd** in the department based on 5th semester GPA.

**Dec 2025**

**Dec 2024**